

Creed v8.1 Wireless Acute DBS

Target journal: Journal of Neuroscience Methods (JNM)

Working Title: A wireless, configurable bilateral deep brain stimulation platform for acute experiments in freely moving rodents

Provisional positioning: Creed v8.1 (**or whatever we call it**) is a configurable wireless constant-current DBS platform designed to remove tethering while retaining the stimulation flexibility and electrical robustness needed for acute rodent experiments.

Central Message: Creed v8.1 is a configurable wireless DBS platform designed to remove tethering while retaining the stimulation flexibility and electrical robustness needed for acute rodent experiments. Electrode impedance and stimulator compliance are treated as a coupled design problem, and the resulting system is validated electrically and in vivo against a conventional tethered approach.

Overall manuscript strategy

- The manuscript follows a methods-centered progression: unmet experimental need, system architecture, electrical and electrode-load characterization, comparison with existing methods, and in-vivo functional validation.
- Creed v8.1 is positioned as an acute wireless research instrument rather than as a power-optimized chronic implant.
- The architecture is presented first as a generalized engineering framework, with Creed v8.1 occupying one point within that design space.
- Electrode impedance and compliance voltage are treated as a coupled systems problem because the usable stimulation envelope depends on both the current source and the electrode-tissue load.
- The biological experiment serves as functional validation of the method rather than as a full DBS efficacy study.
- Prior wireless DBS systems establish context and comparison without turning the manuscript into a review paper.

1. Introduction

Rodent DBS experiments commonly rely on tethered stimulation hardware. Tethers can constrain movement, complicate behavioral assays, and add handling or commutator-related complexity. Existing wireless systems often optimize around chronic implantation, extreme miniaturization, fixed stimulation modes, or highly specialized electronics.

Acute experiments occupy a different engineering space. Real-time control, parameter flexibility, robust constant-current delivery, bilateral stimulation, and generous compliance voltage may be more important than maximizing battery life. Creed v8.1 is presented as a practical implementation of this acute, flexible design space.

The manuscript describes the system architecture, electrical and electrode-load characterization, comparison with existing stimulation approaches, and an in-vivo functional demonstration of wireless DBS.

2. System architecture and implementation

2.1 General DBS architecture

Figure 1 presents a generalized DBS block architecture rather than a detailed Creed schematic. The major functional blocks are:

- Battery / energy source
- Power conditioning and compliance-voltage generation
- Stimulation timing / waveform generation
- Current regulation
- Polarity and output routing
- Electrode interface
- Microcontroller / supervisory logic
- Wireless control and telemetry

The same architecture exposes the main design tradeoffs: battery-direct versus boosted rails, constant-current versus constant-voltage stimulation, available compliance voltage, unipolar versus bipolar supplies, active versus passive charge recovery, wireless control versus autonomous operation, and acute flexibility versus chronic power optimization.

2.2 Creed v8.1 implementation

Creed v8.1 implements this architecture with battery-powered supply generation, positive and negative stimulation rails, a constant-current output stage, bilateral output routing, charge-balancing or recharge circuitry, microcontroller-based stimulation timing, and BLE control for live parameter adjustment.

The implementation is characterized by its programmable stimulation amplitude, pulse width, frequency, channel selection, physical form factor, mass, and practical acute operating duration. Detailed schematics and implementation-specific circuit information can remain in supplementary material while the main text emphasizes the functional architecture.

2.3 Design considerations

The architecture reflects a set of coupled experimental requirements rather than a single universally optimal DBS design. Relevant variables include stimulation current and pulse width, expected electrode impedance, required compliance voltage, waveform and charge-balancing requirements, number of channels, wireless interaction needs, allowable device mass and volume, and acute runtime versus chronic power budget.

The v9 architecture review provides a useful contrast in optimization target: when chronic power consumption becomes dominant, quiescent current, rail generation, charge recovery, and wireless duty cycle become stronger design constraints. This contrast supports the broader architecture discussion without making v9 a subject of the present manuscript.

3. Electrode impedance and stimulation robustness

Electrode characterization is a central part of the methods contribution because the stimulation hardware cannot be evaluated independently of the electrode-tissue load.

3.1 Electrode construction

The common twisted-pair electrode is defined by wire material and diameter, twist geometry, exposed conductive surface, tip preparation, implantation configuration, and fabrication repeatability.

3.2 Bench impedance characterization

Bench measurements characterize impedance magnitude and phase across frequency, electrode-to-electrode variability, and the spread of loads that the stimulator must tolerate. Nyquist representation or equivalent-circuit fitting is included only if it materially clarifies the electrode behavior.

3.3 In-vivo impedance characterization

In-vivo impedance measurements extend the bench characterization to the implanted state, including the planned approximately six-week time point. Where available, bench, implantation-period, and later in-vivo measurements provide a direct view of how the load evolves over time.

(optional) 3.4 Impedance-compliance design relationship

For constant-current stimulation, required output voltage rises with stimulation current and effective electrode-tissue impedance, with additional headroom required by electrode polarization and the current-source circuitry. The measured impedance range therefore defines the compliance voltage needed for robust current regulation.

A design-space figure can map stimulation current and measured impedance against required compliance voltage, with representative supply limits showing where a given architecture remains in regulation and where it saturates.

4. Electrical validation of the wireless stimulator

4.1 Current accuracy and pulse fidelity

Electrical validation covers commanded versus measured stimulation current, pulse-width accuracy, frequency accuracy, representative pulse shape, and repeatability across the expected load range.

4.2 Load tolerance and compliance

Load sweeps establish the point at which the current source can no longer maintain the commanded current. These measurements provide the experimental counterpart to the impedance-compliance analysis and define the usable stimulation envelope of the device.

4.3 Wireless control

Wireless validation demonstrates reliable stimulation initiation and live parameter changes over the spatial range needed for behavioral experiments. The emphasis remains on experimental utility rather than communications benchmarking.

4.4 Battery and acute operating duration

Representative active current consumption and practical experiment duration establish that the device supports the intended acute use case. Runtime is interpreted relative to experimental need rather than against chronic implant requirements.

5. Comparison with existing methods

The comparison situates Creed v8.1 relative to conventional tethered stimulation and previously published wireless rodent DBS systems. Relevant dimensions include device size and mass, number of stimulation channels, constant-current capability, compliance voltage or supported load range, pulse amplitude, pulse-width and frequency flexibility, wireless programmability, intended operating duration, and availability of reproducible hardware or software resources.

The comparison emphasizes where Creed sits within the design space rather than claiming universal superiority. Its intended advantage is the combination of untethered operation, flexible stimulation control, and electrical headroom for acute rodent experiments.

6. In-vivo functional validation

6.1 Experimental paradigm

The in-vivo study provides a compact validation of practical function. The core comparison can include baseline or DBS-off behavior, standard tethered DBS, and Creed wireless DBS, depending on the final experimental design.

6.2 Video-based behavioral analysis

Video-derived movement metrics provide an objective behavioral readout. Candidate measures include body-centroid velocity, distance traveled, fraction of time moving, movement-bout frequency, and movement-bout duration.

6.3 Functional equivalence / validation

The primary question is whether Creed produces the expected behavioral effect in the same direction and approximate range as a standard tethered method. The result establishes functional equivalence of the stimulation method without extending the claim to a full biological efficacy study.

7. Discussion

Creed v8.1 provides flexible wireless DBS for acute or short-duration freely moving experiments. Its design prioritizes stimulation flexibility, electrical headroom, bilateral operation, and live wireless control.

The electrode measurements reinforce that DBS performance is a systems problem: electrode impedance and stimulator compliance jointly determine whether commanded constant-current stimulation can actually be delivered.

Acute and chronic systems therefore occupy different optimization spaces. A chronic architecture may reasonably favor lower quiescent current, reduced supply complexity, alternative charge-recovery approaches, and more aggressive wireless duty cycling, whereas the v8.1 architecture favors experimental flexibility and robust stimulation.

Key limitations include device mass and volume, battery life, continuously powered circuitry, BLE power requirements, and the limited scale of the in-vivo validation. Future development can extend the platform toward lower-power chronic derivatives without changing the role of v8.1 as the validated acute system.

Proposed figure set

- **Figure 1:** General DBS architecture and design tradeoffs, with Creed v8.1 shown as one implementation.
- **Figure 2:** Twisted-pair electrode construction and bench / in-vivo impedance characterization.
- **Figure 3:** Impedance x stimulation current x compliance-voltage design space.
- **Figure 4:** Bench electrical performance including current accuracy, pulse fidelity, and load/compliance testing.
- **Figure 5:** In-vivo setup and wireless-versus-tethered behavioral validation.

Supplementary material candidates

- Detailed circuit schematic
- PCB layout and fabrication files
- Bill of materials
- Firmware and BLE protocol details
- Extended waveform validation
- Additional impedance measurements
- Mechanical drawings or enclosure files
- Behavioral analysis code

JNM fit and scope considerations

- Novelty is centered on the method and the capability it enables, rather than an isolated component-level invention.
- The generalized architecture gives the paper transferable engineering value.
- Quantitative comparison with existing methods establishes what is new and when the platform is preferable.
- Bench characterization defines the operating envelope under realistic loads.
- The in-vivo experiment demonstrates practical neuroscience utility without requiring a full biological study.
- Reproducible hardware, firmware, analysis code, and fabrication information strengthen the methods contribution where they can be released.
- The acute-use framing makes the v8.1 design tradeoffs coherent without implying that chronic power efficiency is a design goal.

Scope boundaries

- Avoid a comprehensive review of all DBS stimulator architectures.
- Avoid detailing the full development history of earlier Creed revisions.
- Keep discussion of the proposed v9 chronic architecture limited.
- Avoid mechanistic interpretation of the behavioral effect.
- Avoid large-scale efficacy claims from a small validation cohort.
- Limit wireless-range benchmarking to what is needed to establish experimental utility